

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

C02: *Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)*

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I _____V.Lasya(192524186)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.lasya

Debugging & Optimisation Task

1. Smart Office WLAN/WWAN Connectivity Issues

A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than 1,200 employees use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's Wireless Local Area Network (WLAN). The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use 4G/5G Wireless Wide Area Network (WWAN) services to access the company's servers, customer databases, and business applications from remote locations.

During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

Answer:

1. Root Causes Identified:

- Slow speeds during peak hours are caused by too many devices (1,200+) sharing a limited number of access points (AP overloading/channel contention).
- Frequent disconnections while moving between floors indicate poor roaming configuration — access points are likely using different SSIDs/channels without seamless handoff (no 802.11r/k/v fast roaming).

- Weak signals in conference rooms point to poor AP placement, signal attenuation from walls/floors, and insufficient AP density in high-usage zones.
- Video conferencing freezes are due to insufficient bandwidth prioritization — no Quality of Service (QoS) policy for real-time traffic.
- Uneven AP utilization shows a lack of load balancing / band steering between 2.4 GHz and 5 GHz radios.
- WWAN dropouts in rural/highway areas result from weak cellular tower coverage and lack of failover to alternate carriers.
- Outdated security protocols (e.g., WEP/WPA) on some devices create vulnerabilities to eavesdropping and unauthorized access.

2. Debugging Steps:

- Conduct a wireless site survey (heat-map analysis) across all six floors to measure signal strength (RSSI), interference, and channel overlap.
- Analyze AP logs and WLAN controller statistics to identify overloaded APs, client counts per AP, and retransmission rates.
- Use packet capture/protocol analyzers (e.g., Wireshark) during video conferencing sessions to check for packet loss, jitter, and delay.
- Audit all connected devices to identify those still using outdated security protocols (WEP/WPA/TKIP).
- Review WWAN signal logs for field engineers to map dead zones along travel routes.

3. Optimization Recommendations:

- Increase AP density and reposition APs based on the heat-map survey to eliminate coverage gaps in conference rooms and floor edges.
- Enable seamless roaming (802.11r/k/v) with a unified SSID across all floors for uninterrupted handoff.
- Implement band steering and load balancing to evenly distribute clients between 2.4 GHz and 5 GHz/6 GHz bands.
- Apply QoS (WMM) policies to prioritize video conferencing and VoIP traffic over bulk downloads.

- Upgrade all devices to WPA3 security and enforce 802.1X authentication for enterprise-grade protection.
- For WWAN, deploy dual-SIM/multi-carrier failover and use 5G where available to improve reliability for field engineers in remote areas.

2. ATM Network Performance Optimization

A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an Asynchronous Transfer Mode (ATM) network to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size 53-byte cells and establishes Virtual Circuits (VCs) to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.

Answer:

1. Faults Identified:

- High switch utilization during peak hours indicates insufficient trunk capacity or poor traffic distribution across ATM switches.
- Congested Virtual Circuits (VCs) show that bandwidth is not reserved appropriately for high-priority traffic.
- Jitter and packet loss in VoIP calls suggest that voice traffic is not assigned to the correct low-latency service category.
- Applications using inappropriate ATM service categories (e.g., real-time traffic mapped to UBR instead of CBR/rt-VBR) cause increased delay.
- Uneven bandwidth allocation among traffic classes points to a missing or misconfigured traffic contract/admission control policy.

2. Debugging Steps:

- Monitor per-VC statistics (cell loss ratio, cell delay variation) at each ATM switch to identify congested circuits.
- Check the ATM Adaptation Layer (AAL) mapping to confirm which service category (CBR, rt-VBR, nrt-VBR, ABR, UBR) is assigned to each application.
- Review Connection Admission Control (CAC) logs to see whether new VCs are being admitted despite insufficient available bandwidth.
- Use traffic analyzers to measure cell delay variation (CDV) and cell loss ratio (CLR) for voice/video circuits during peak periods.

3. Optimization Recommendations:

- Reassign real-time voice and video traffic to CBR (Constant Bit Rate) or rt-VBR (real-time Variable Bit Rate) service categories to guarantee low delay and jitter.
- Assign bulk/non-critical data transfers to UBR or ABR categories so they do not compete with real-time traffic.
- Implement stricter Connection Admission Control (CAC) to prevent oversubscription of congested VCs.
- Use traffic shaping and policing at the network edge to smooth bursty traffic before it enters the ATM backbone.
- Add additional trunk capacity or reroute some VCs to underutilized switches to balance the load across the network.
- Continuously monitor QoS parameters (CLR, CDV, CTD) and dynamically re-negotiate traffic contracts as usage patterns change.

3. MPLS Network Fault Debugging and Optimization

A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a Multiprotocol Label Switching (MPLS) network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences.

Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

Answer:

1. Faults Identified:

- Congested Label Switched Paths (LSPs) alongside underutilized paths show a lack of traffic engineering (TE), causing all traffic to follow the shortest IGP path instead of being distributed.
- Incorrect label mappings on several routers cause packets to be forwarded on wrong or suboptimal paths, increasing latency and causing loops or drops.
- Poor voice/video quality indicates no traffic prioritization (no MPLS QoS/Diffserv-aware TE) for real-time applications.
- Uneven bandwidth utilization across the backbone confirms that Constraint-Based Routing (CB-TE) has not been enabled.

2. Debugging Steps:

- Verify the Label Forwarding Information Base (LFIB) on each Label Switch Router (LSR) to check for incorrect or stale label bindings.
- Use MPLS ping/traceroute (LSP ping) to test the health and path of each Label Switched Path and identify broken or misrouted LSPs.
- Check LDP (Label Distribution Protocol) or RSVP-TE session status between routers to confirm labels are being correctly advertised and bound.
- Analyze bandwidth utilization per LSP using network monitoring tools (e.g., SNMP/NetFlow) to identify congested versus idle paths.
- Review the traffic engineering database (TED) to confirm whether link bandwidth and constraints are correctly advertised via OSPF-TE/ISIS-TE.

3. Optimization Recommendations:

- Correct the label bindings on affected routers and refresh the LFIB tables to restore accurate label-based forwarding.
- Enable RSVP-TE with Constraint-Based Routing so traffic is distributed across multiple LSPs based on available bandwidth, not just shortest path.
- Configure MPLS Traffic Engineering (MPLS-TE) tunnels with explicit paths to route around congested links and utilize idle capacity.
- Apply Diffserv-aware Traffic Engineering (DS-TE) to prioritize VoIP and video conferencing traffic with guaranteed bandwidth classes.
- Implement Fast Reroute (FRR) for automatic failover in case an LSP fails, minimizing downtime for critical applications.
- Continuously monitor LSP utilization and use automatic bandwidth adjustment (Auto-Bandwidth) to adapt to changing traffic patterns.

4. Data Transmission Error Control Debugging and Optimization

A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.

Answer:

1. Faults Identified:

- Undetected corrupted frames indicate that the current error detection mechanism (likely a simple parity check or weak checksum) is too weak to catch multi-bit or burst errors.

- Electrical interference and signal attenuation near scanners/robots cause burst errors typical of noisy industrial environments (motors, RF interference from RFID readers).
- Network congestion during peak hours contributes to buffer overflow drops, which can appear as missing or corrupted records.
- Since only some corrupted frames are retransmitted, the Automatic Repeat reQuest (ARQ) mechanism is not consistently triggered — pointing to unreliable or absent CRC verification at the receiver.

2. Debugging Steps:

- Capture and inspect frames at the receiver to determine the error detection method currently in use (parity, checksum, or CRC) and its effectiveness.
- Measure the Bit Error Rate (BER) on affected links during peak hours versus off-peak hours to correlate errors with interference/congestion.
- Check whether retransmission (ARQ) is enabled and functioning for all frame types, or only for a subset of traffic.
- Inspect physical layer conditions (cable shielding, distance from motors/robots, RFID reader placement) for sources of electromagnetic interference.

3. Optimization Recommendations:

- Replace weak parity/checksum methods with a strong Cyclic Redundancy Check (CRC-32), which reliably detects burst errors common in industrial environments.
- Implement Forward Error Correction (FEC) such as Hamming code or Reed–Solomon coding so common errors can be corrected without requiring retransmission, reducing delay.
- Use Hybrid ARQ (combining CRC detection with automatic retransmission) to ensure every corrupted frame is either corrected or retransmitted reliably.
- Improve physical layer conditions: use shielded/twisted-pair or fiber cabling, relocate cabling away from motors and RFID transmitters, and add repeaters where attenuation is high.
- Apply traffic prioritization/QoS and increase buffer capacity at congestion points to reduce peak-hour packet drops.
- Add periodic data reconciliation (checksums at the database/application layer) to catch and correct any residual undetected errors in shipment/inventory records.

5. Online Examination Platform Flow Control Debugging and Optimization

A leading online education platform conducts live classes and online examinations for more than 15,000 students across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed 10 Gbps connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

Answer:

1. Faults Identified:

- Receiver buffer overflow and application crashes show that the server is sending data faster than slower client devices can process — classic sender/receiver speed mismatch.
- This indicates the system is likely using a fixed-rate or stop-and-wait style transmission with no proper sliding window / dynamic flow control.
- Duplicate packets and excessive retransmissions suggest that timeout values are too short relative to actual round-trip time, causing premature retransmission before acknowledgements arrive.
- Reduced throughput despite a 10 Gbps server link confirms the bottleneck is at the client/receiver side and last-mile network, not the server capacity.

2. Debugging Steps:

- Capture traffic at both server and client ends to compare send rate versus the receiver's actual processing/acknowledgement rate.
- Analyze TCP window size negotiation and check whether receive window (RWND) values are being respected by the sender.

- Measure round-trip time (RTT) variability across different student connections (broadband vs mobile) to identify why timeouts trigger unnecessary retransmissions.
- Review server logs for buffer overflow events and correlate them with the volume/size of simultaneous data pushes (question papers, images, videos).

3. Optimization Recommendations:

- Implement a proper sliding window flow control mechanism so the server dynamically adjusts transmission rate based on each client's advertised receive window.
- Enable adaptive/dynamic window sizing (e.g., TCP window scaling) instead of a fixed transmission rate, allowing slower clients to signal reduced capacity.
- Set retransmission timeout (RTO) values adaptively based on measured RTT for each connection to avoid unnecessary duplicate retransmissions.
- Use content delivery optimization such as adaptive bitrate streaming for videos and compressed/paginated content delivery for question papers to reduce load per client.
- Apply congestion control algorithms (e.g., TCP Cubic/BBR) alongside flow control to balance throughput fairly among 15,000+ simultaneous connections.
- Introduce load balancing/CDN edge servers to distribute simultaneous downloads geographically, reducing last-mile congestion during peak examination periods.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation